

AN OPTIMAL CONTROL ANALYSIS OF SAN FRANCISCO'S MUNI LIGHT RAIL

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ABSTRACT. In this paper, we formulate and solve an optimal control problem for San Francisco's Muni Metro light rail. Our problem models the acceleration and braking as a single control, optimizing for energy cost, brake wear, and smoothness of the control function. We present solutions for two cases: time as an independent variable and time as a dependent variable. For the former case, we find the optimal control function to move a light rail to a predefined end position and end time. This scenario covers most Muni travel cases since light rails adhere to a strict timetable and the distance between stations is fixed. Our simulated routes include those that are straight, curved, and have grades. We then modify the model to make time the dependent variable to account for the situation where a light rail approaches a terminal station, with no train blocking the station ahead. We conclude with a sensitivity analysis on uncertain parameters and robustness checks using varying grade profiles.

Keywords. Optimal Control, Constrained Optimization, Mathematical Modeling
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1. INTRODUCTION

San Francisco's Muni Metro (Muni) is a light rail system that serves the city proper, with an average of 96,000 boardings per day in 2025 and a total of 29 million passengers in 2024 [23]. Muni's lines are extensive, with lines running to the city's western border and as far south as the Daly City border. Like many U.S. transit systems, Muni is designed as a spoke-hub system [16], engineered to funnel commuters from the city's outer neighborhoods to the downtown business district. San Francisco (S.F.) is one of the few cities in the U.S. where a sizable share of the population commutes by public transportation [21]. Roughly 17% of workers commute by public transit, fourth overall, highlighting Muni's significance to the city's residents.

Determining the solution to an optimal control problem for Muni is significant for a variety of reasons. Transit serves an important role in connecting residents and generating economic activity [18]. The solution can be used to program driver assistance software, such as cruise control. It can also be generalized to autonomous people mover systems for simpler, more controlled routes. Lastly, the solution provides valuable information to drivers via optimal driving strategies for difficult routes, thus ensuring passenger comfort and safety.

Our model builds on the one developed by Udriste [20], which highlighted the importance of optimizing energy consumption. In addition to energy cost, our model includes brake wear cost, passenger discomfort penalty, and regenerative braking to mimic Muni's Siemens LRV4 electric light rails. We simulate our model on train routes that include grades and curvatures to match S.F.'s geography. For the remainder of this paper, we use the terms light rail and train synonymously. Section 3.1-3.2 analyzes the optimal trajectory with time as the independent variable, while Section 3.3 repeats the analysis with time as the dependent variable.

2. DESCRIPTION OF THE MODEL

From a passenger’s perspective, arriving earlier is better. However, in transit systems, the aim is to adhere to the timetable. This ensures proper transfer connections, maintains passenger expectations, and prevents bunching up [6]. Even if a train can arrive early, it may still have to wait for the train ahead to exit the station before it can pull in, which is the case for most Muni stations. As such, we first model the case where the travel time T is fixed. We express the model as a system of ODEs:

$$x(t) = \begin{bmatrix} p(t) \\ v(t) \end{bmatrix}, \quad \dot{x}(t) = \begin{bmatrix} \dot{p}(t) \\ \dot{v}(t) \end{bmatrix} = \begin{bmatrix} v(t) \\ \frac{1}{m} \left(u(t) - F_{\text{drag}}(v) - F_{\text{roll}}(v) \right) \end{bmatrix}. \quad (2.1)$$

The two friction forces, air drag, and rolling friction, are defined as

$$F_{\text{drag}}(v) = c_{\text{drag}} v(t)^2; \quad F_{\text{roll}}(v) = c_{\text{roll}} m g \cdot \text{sgn}(v(t)). \quad (2.2)$$

$x(t)$ is the state vector, consisting of position $p(t)$ and velocity $v(t)$. The time derivative of position is $v(t)$, and the time derivative of $v(t)$ is acceleration. m refers to mass and g refers to the gravitational acceleration constant. The evolution of the system depends on both the current state and the control input $u(t)$, measured in Newtons. The acceleration term comes from Newton’s Second Law. Force is composed of $u(t)$, which controls the amount of force applied during acceleration and deceleration, or braking. We include an air drag force and a rolling friction force. The drag coefficient, c_{drag} , is given by the formula $c_{\text{drag}} = 0.12 + 0.0075 * L$ from Hara [10], where L is the length of the train. Although the formula is intended for streamlined trains, we assume LRV4s are streamlined enough to make this a close approximation. From Siemens’ spec sheet, we get $L = 22.86$ m and $m = 34473$ kg [19]. Plugging in for L , we get $c_{\text{drag}} = 0.29$. $c_{\text{roll}} = 0.001$ was gleaned from Maanshan’s report [13]. Equation 2.2 includes $v(t)^2$ because we’re using a simplified form of the drag equation [22].

The total cost function is defined as

$$J(u, x) = 4,880 \|x(T) - x_f\|^2 + \int_0^T l_{\text{energy}}(u(t), v(t)) + l_{\text{jerk}}(u(t)) \, dt. \quad (2.3)$$

The first term is the terminal state penalty, where $X(T)$ is the model’s final state vector and x_f is the target state vector. We set the terminal error to be relatively costly because (1) the train should not be moving at the terminal state and (2) Muni’s accessibility ramps must align precisely with the train doors; otherwise, misalignment could result in legal action. $40,000 \cdot 0.122 = 4,880$ is the product of the average ADA lawsuit cost and proportion of the U.S. population with mobility disability [4][2]. This represents the expected penalty from state deviation. Sources give a wide range of ADA lawsuit settlement amounts. As an approximation, we use a value on the lower end of the range because not everyone sues. Deviation from the terminal state is quadratic, capturing the notion that large deviations increase the likelihood of a lawsuit, while small deviations are less likely.

For the energy terms, measured in joules, we have

$$l_{\text{energy}}(u(t), v(t)) = \begin{cases} C_e u(t) v(t), & u(t) \geq 0, \\ \eta_{\text{regen}} C_e u(t) v(t) + C_w |u(t)|, & u(t) < 0 \end{cases}. \quad (2.4)$$

$l_{\text{energy}}(t)$ handles two cases: acceleration and deceleration. C_e is the cost of energy in dollars per joule, set to S.F.’s average electricity price [5], and C_w is the brake wear cost. The wear cost calculation is lengthy and not especially important to the objective function; for brevity’s sake, the calculation is placed in the Appendix (6.1). In short, the wear cost is derived from the cost of a brake divided by the impulse on a brake during stopping. The

model assumes no engine wear because engines typically last the lifetime of a light rail. $\eta = 0.3$ is the proportion of energy recouped by regenerative braking. It is set to the value given in a paper by Khodaparastan [12].

The last term in the time integral is the penalty for sharp changes in $u(t)$,

$$l_{jerk}(u(t)) = R_{loss}(u(t)) \frac{d^2 u}{dt^2} = (1 - \beta(a(t)))(n_{pass} \cdot f) \frac{d^2 u}{dt^2}. \quad (2.5)$$

Large changes in the second derivative of $u(t)$ are penalized by revenue loss. Revenue loss is computed as the product of fare, number of passengers, and proportion of uncomfortable passengers for a given acceleration. The range of comfortable acceleration is 0.9 m s^{-2} to 1.47 m s^{-2} [1]. We model this as a function where 100% of passengers are comfortable within the range, but the proportion decays linearly as acceleration deviates from the range (Equation 2.6, Appendix 6.3 for figure). We have 0 passengers comfortable at 0 m s^{-2} and past 2.37 m s^{-2} ($0.9 + 1.47 = 2.37$ to make the function symmetric). When acceleration is too fast, riders are uncomfortable. If the acceleration is too slow, the train underperforms and feels sluggish to passengers.

$$\beta(a(t)) = \begin{cases} \frac{a}{0.9} & \text{if } 0 < a < 0.9 \\ 1 & \text{if } 0.9 \leq a \leq 1.47 \\ 1 - \frac{a-1.47}{0.9} & \text{if } 1.47 < a < 2.37 \\ 0 & \text{otherwise} \end{cases}. \quad (2.6)$$

For $t \in [0, T]$, and a predefined final state x_f , the optimization problem is

$$\min_{u(t)} J(u, x) \quad (2.7)$$

subject to

$$\begin{aligned} 0 &\leq v_t \leq v_{max} \\ p_f - 5 &\leq p(T) \leq p_f + 5 \\ v_T &= 0 \\ -7.909 \times 10^4 &\leq u(t) \leq 6.793 \times 10^4. \end{aligned}$$

The constraints disallow the train from moving backward and from exceeding the operational speed limit v_{max} , set to 22.36 m s^{-1} , as given in the Siemens spec sheet [19]. The terminal position must be within 5 m of the intended position, and the final velocity must be 0. Margin of error for the position was introduced because algorithmic convergence was proving difficult with a strict constraint. To circumvent the issue, we opted for a high terminal penalty rather than a strict constraint to ensure the correct terminal position. Lastly, we bound the control function between $-7.909 \times 10^4 \text{ N}$ and $6.793 \times 10^4 \text{ N}$ [19].

3. ANALYSIS

3.1. Optimal $u(t)$ for Time-Independent Model. With the optimal control problem specified, we now solve the optimization problem. For the entire analysis section, the initial condition is set to position 0 and velocity 0. We set a target travel distance d , measured using Google Maps [9] or TrailRouter [17]. Velocity at the terminal state is 0.

$$x(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_f = \begin{bmatrix} d \\ 0 \end{bmatrix} \quad (3.1)$$

The time step was chosen such that $dt \propto T$. If T is large, we take larger time steps, and vice versa. Although the model is a continuous function of t , we discretize the continuous interval $[t, T]$ into k steps of size dt , from $[0, N]$, where $N = \frac{T}{dt}$, to numerically solve. The

number of passengers is set to half capacity, with a 50/50 split between men and women. We use the average U.S. weight for each sex given by the CDC [8].

Figure 1 shows symmetric trajectories for $u(t)$, $v(t)$, and $p(t)$, for the straight route case. The optimal trajectory is to induce a large amount of power, then linearly reduce. At $\frac{T}{2}$, the trajectory switches from accelerating to braking. Braking is then applied linearly until a complete stop. Velocity and power are not at the boundary; the train is not being pushed to its limit.

Even with large power, since the train is heavy, the velocity doesn't abruptly change. We see velocity increases from 0 to 5 m s^{-1} in 10 seconds, which is a very gradual acceleration. The position graph shows that the terminal condition is satisfied with a velocity of 0 and a position of 740 m.

3.2. Optimal $u(t)$ for Time-Independent Model with Grade & Curvature. We now analyze scenarios where the train's route includes curvature and grade. It is important to analyze these situations because, realistically, trains don't always travel on straight, flat routes. Grade, in particular, is important to model because of San Francisco's hilly terrain.

For the curved trajectory, we modify the optimization problem $J(u, x)$ to include a lateral acceleration constraint (Equation 3.2) set to 1.47 m s^{-2} , which is the end of the range given by Bae [1].

$$a_{lat}(t) = \frac{v(t)^2}{R(t)} \leq a_{lat}^{(max)} = 1.47. \quad (3.2)$$

$R(t)$ is a time-varying curve radius, measured in meters. We acknowledge that, instead of a constraint, a soft penalty could have been used, but we wanted to illustrate the change in behavior during a turn.

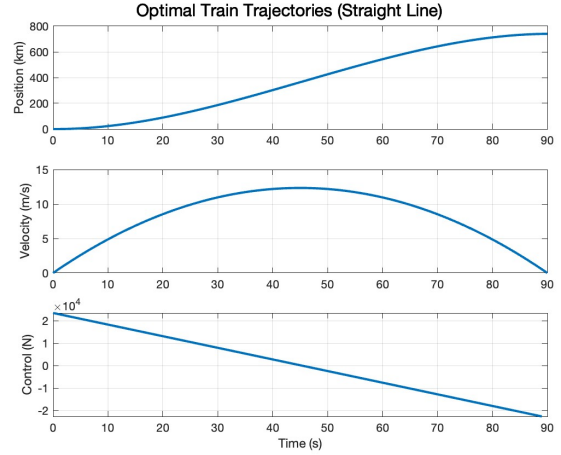
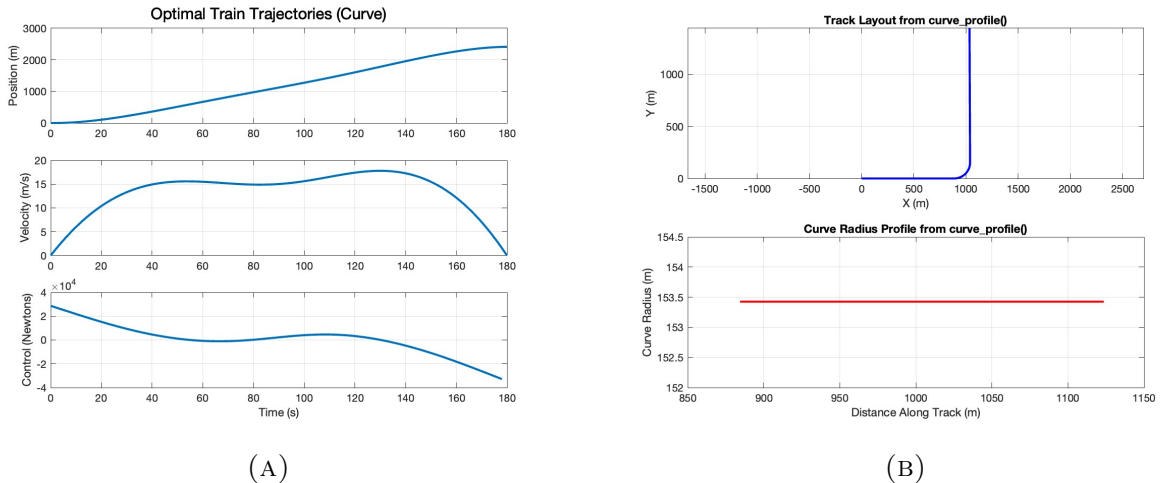


FIGURE 1. Trajectory Straight Line



(A)

(B)

FIGURE 2. Optimal Trajectories for Curved Path

For curved track simulation, we set $d = 2414\text{m}$ and $T = 3\text{min}$, simulating the route from Embarcadero & Folsom to Powell Station. The real travel time is 6 min, but it was reduced to 3 min to account for skipping two stations in the simulation. The turn is 90° , starting 884m into the trip, ending at 1125m, giving an arc length of 241 m. The results shown in Figure 2(A) are similar to the results of Section 3.1. However, the key difference is that $u(t)$ hovers around 0 during the turn, attempting to maintain constant velocity and zero acceleration. Looking at the overall trajectory, a large amount of power is first used to overcome inertia, keeping acceleration smooth. Power is gradually reduced approaching the turn, and after the turn, the brake strength is gradually increased to stop the train. The result indicates the model is capable of handling small turns. The lateral acceleration constraint does what it was designed to do: keep the train from going too fast during a turn. Figure 2(B) shows the travel route.

The last scenario for the time-independent model is to simulate a route with a grade. We remove the lateral acceleration term and replace it with a gravity term:

$$F_{hill} = mg \sin(\theta) = mg \sin(\arctan(\delta)). \quad (3.3)$$

We use $\arctan$ because $\tan \theta = \delta \rightarrow \theta = \arctan(\delta)$, where $\delta = \frac{\Delta h}{\Delta x}$ is the grade (rise over run). In this simulation, we aim to mimic the route from Church & Market to Church & 24th. Using elevation data from topology maps [7] and grade information from Wikipedia [24], we recreated the route, shown in Figure 3(B). The peak is around 29 m, and the elevation at Church & 24th is around 24 m. The simulated hill profile is relatively close. Normal travel time between the two stations is 8 min, but the simulated travel time was reduced to 3 min because the simulation ignores traffic lights, stop signs, and the six stops in between. Distance is set to 1818 m, and emergency braking is permitted for this scenario.

The optimal trajectory gradually increases the amount of power into the train. When nearing the peak, power is reduced, and the rest of the slope is used to slow down the train. At the peak ($t = 120\text{s}$, 1100 m), brakes are applied but not at the emergency brake level. Overall, the trajectory is smooth, even with no smoothing terms used. One caveat is that the hill profile is not entirely continuous. It is pieced together, so there exist points where the derivative does not exist. This may cause artifacts in the control trajectory, but we do not observe any here.

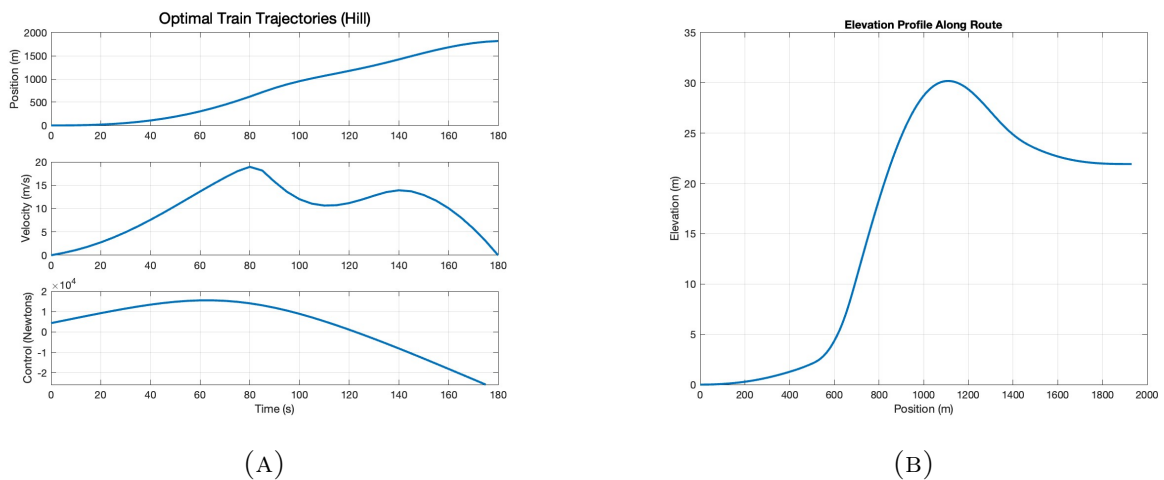


FIGURE 3. Optimal Trajectories for Grade/Hill

3.3. Optimal $u(t)$ for Time-dependent Model. Using the chain rule, we rewrite the system of ODEs to have time t as the dependent variable and position p as the independent

variable:

$$x(p) = \begin{bmatrix} t(p) \\ v(p) \end{bmatrix}, \quad \frac{d}{dp} \begin{bmatrix} t(p) \\ v(p) \end{bmatrix} = \begin{bmatrix} \frac{1}{v(p)} \\ \frac{u(p) - F_{drag}(p) - F_{roll}(p)}{m v(p)} \end{bmatrix}. \quad (3.4)$$

State is now a vector of time and velocity, both functions of position. The first differential equation describes how time changes with respect to position. Intuitively, the faster the train is moving, the less time it takes to travel a given distance. The second equation describes how velocity changes with respect to position. When the train is moving slowly, it takes more time to traverse each meter. As a result, the applied force acts over a longer duration per meter, producing a larger change in velocity per meter. In contrast, if the train is moving quickly, it covers each meter almost instantly, so the same force acts for less time per meter, producing a smaller change in velocity per meter.

The objective function remains the same (Equation 3.5), except that the independent variable is now p . We add a penalty term for the second derivative to encourage smoothness in $u(t)$, modulated by parameter λ . The smoothing term is, admittedly, arbitrary; therefore, we'll conduct a sensitivity analysis on this parameter to ensure results are intuitive (Section 3.4).

$$\min J(u, x) = 4,880 \|x(P) - x_f\|^2 + \int_0^P l_{energy}(p) + l_{jerk}(p) + \lambda \left(\frac{du}{dp} \right)^2 dp. \quad (3.5)$$

s.t.

$$v(P) = 0; \quad 0 \leq v(p) \leq v_{max}$$

The constraints are simpler than before; there's no need to put a constraint on the final position. The significance of this model is that it allows us to find the optimal travel time T , instead of having to predefine it. For a travel distance of 740 m, with a straight route, the optimal travel time was found to be 1.54 min. From Figure 4, we see that the trajectories are very similar to Figure 1. The one difference is that this model has sharper acceleration and deceleration.

3.4. Sensitivity Analysis. For our sensitivity analysis, we test three cases. We first examine how the number of passengers influences the optimal trajectory. We then test how the time-dependent model's trajectory changes with λ . Lastly, we stress test the train on different hill profiles, keeping all other parameters the same as before.

For passenger count sensitivity, we vary the passenger count from 1 to 193. 193 is the maximum capacity of a Muni train, as given in the Siemens spec sheets [19]. We again assume an even split of men and women for the weight and use the straight route, time-independent model. Figure 4(B) shows that velocity and position trajectories are nearly identical across passenger counts. The $u(t)$ curves vary by weight because more energy is needed to move a larger mass, due to inertia. For lighter vehicles, less extreme acceleration and braking are required. All $u(t)$ trajectories are linear and switch from acceleration to braking at $\frac{T}{2}$.

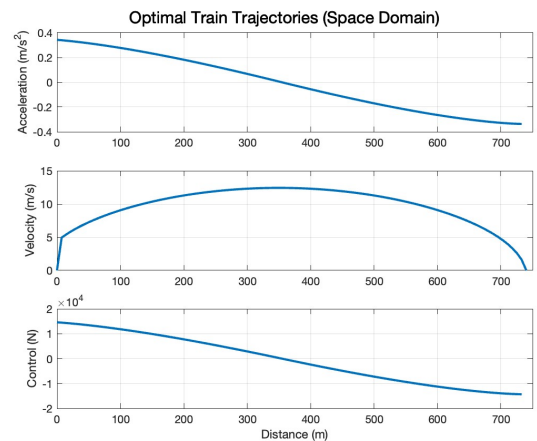
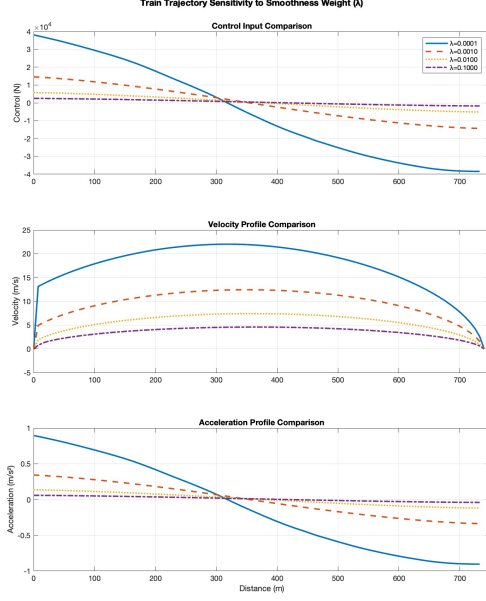
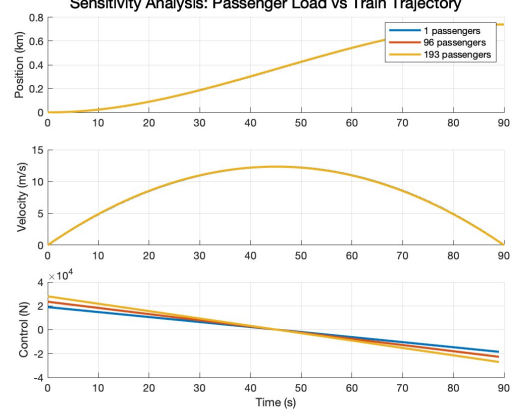


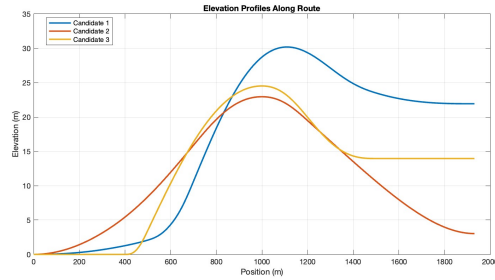
FIGURE 4. Trajectories for Time Independent Model

(A) Sensitivity to λ 

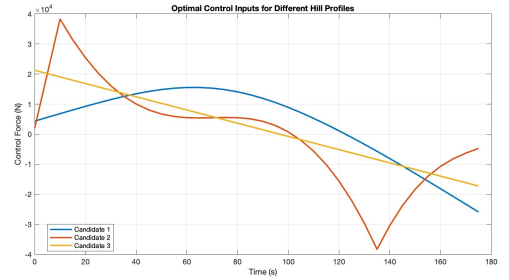
(B) Sensitivity to Number of Passengers

FIGURE 5. Comparison of optimal train trajectories and sensitivity analysis.

Next, we examine the sensitivity of trajectories to λ because it was the most uncertain of the parameters in the time-dependent model. The λ values were spaced on the log scale. As the smoothing parameter increases, the trajectories opt for less power and slower arrival times, reflecting the tradeoff between earliness and smoothness. Figure 5(A) shows the trajectories, and Appendix 6.2 contains a table of optimal travel time for each λ . The smoothness penalty seems to have a large influence on the trajectory, highlighting a weakness in the time-dependent model. Other cost factors don't seem to make a huge difference in the optimal trajectory.



(A)



(B)

FIGURE 6. Optimal Trajectories for Different Hills

Next, we stress test the train with three hill profiles, each designed to be different. Candidate 1 is the same hill used in Section 3.2, serving as a basis of comparison. Candidates 2 and 3 are not based on any real S.F. route. Instead, candidate 2 is used to test a steep incline and decline. Candidate 3 tests for sharper changes in grade, to see if kink points affect numerical stability. Note that Muni trains are not capable of riding on grades that exceed 10° ; we do not test these scenarios. For candidate 3, the solution is linear. $u(t)$ takes advantage of the flat starting portion to gather as much momentum as possible to overcome the hill, then braking is applied. Sharp changes in grade don't seem to affect the

solution. The trajectory of candidate 1 uses more power than candidate 2 because there is no flat portion to gain speed in the beginning, and no flat portion to decrease speed at the end. For candidate 3, a huge amount of power is induced because the grade during ascent and descent is the largest. The power used is large, but it is not at the boundary. The symmetric optimal control seems to reflect the symmetry in the hill profile.

4. DISCUSSION

Prior papers have proposed optimal control problems and corresponding solutions for trains, but few have applied them to Muni. In this paper, we utilized extensive research to pin down the parameters to create a realistic model of the Muni train, with corresponding realistic routes. Our optimal control model spans two models: time-independent and time-dependent. The time-independent model simulates the case where trains adhere to a timetable to prevent bunching. The time-dependent model covers the case where the route ends at a terminal station, making early arrival possible. The strength of the model is its specificity to S.F. and its ability to capture different route types in the city.

In terms of weaknesses, routes that blend grade and curvature were not tested. We also did not test passenger loads for different hill or curvature profiles. Modifications to penalties and constraints may lead to different results. Some constraints may be overly strict, such as the lateral acceleration constraint, which may be better modeled as a penalty. Our model assumed engine wear was 0, which may not be the case. Lastly, the time-dependent model was found to be quite sensitive to the λ parameter. Ideally, the model should be rewritten to remove this parameter.

Despite the weaknesses mentioned, the model is still useful because it provides insights that weren't obvious from the start. A state penalty matrix was not needed to solve this problem. The optimal trajectory was shown to be symmetric in the straight line case. Given that engine wear wasn't a cost, it was expected that the optimal trajectory would start at high power and coast to save on brake wear, but regenerative braking played a major role in modulating the optimal trajectory. For the hill case, the optimal trajectory decreases power just before the peak and uses the remaining slope to reduce the speed of the train, avoiding the need for heavy braking. The switch from acceleration to braking was found to occur at the peak. The initial hypothesis was that power would start small and increase as it goes up the hill, but the trajectory was more adept than that, taking advantage of any flat region to build momentum. Finally, the time-dependent model provides us with the ability to find the optimal travel time T in cases where a train does not need to adhere to a timetable.

5. CONCLUSION

Public transit is an important service because it provides different options of travel for citizens. In cities where there is no public transit, individuals are forced to opt for private transportation, creating a barrier for low-income or young individuals. In this paper, we focused the scope of our optimal control problem on S.F.'s Muni and simulated real-world routes between stations, including those with grade and curvature. We framed the problem as a constrained optimization problem with a terminal cost and a cost accrued over a trajectory. We started with a time-independent model and redefined the objective function to be time-dependent to capture the edge case where a train is arriving at a terminal station, allowing us to find the optimal T . The strength of this paper is its comprehensiveness, simulating the most important route scenarios in a Muni's journey. Modifications to the time-dependent model to remove the smoothness parameter, additional cost for engine wear, and more comprehensive sensitivity analysis remain as future extensions to the model.

REFERENCES

1. Bae, I., Moon, J., & Seo, J. (2019). Toward a comfortable driving experience for a Self-Driving shuttle bus. *Electronics*, 8(9), 943. <https://doi.org/10.3390/electronics8090943>
2. Curtis, J. (2025, April 21). Understanding ADA Lawsuits and Settlement Amounts. AudioEye. Retrieved August 6, 2025, from <https://www.audioeye.com/post/ada-lawsuit-settlement-amounts/>
3. OpenAI. (2025). ChatGPT (August 8 version) [Large language model]. <https://chat.openai.com/chat>¹
4. Disability impacts all of us Infographic. (2025, April 14). Disability and Health. <https://www.cdc.gov/disability-and-health/articles-documents/disability-impacts-all-of-us-infographic.html>
5. Electricity cost in San Francisco, CA: 2025 Electric rates — EnergySage. (2025, August 3). EnergySage. <https://www.energysage.com/local-data/electricity-cost/ca/san-francisco-county/san-francisco/>
6. Elevated Standards for Bus Operations – Scheduling and Dispatching. (2019). In <https://transitcenter.org>. Sam Schwartz Engineering, D.P.C. Retrieved August 5, 2025, from https://transitcenter.org/wp-content/uploads/2020/05/SSE_bus_reliability_paper.pdf
7. Elevation Contours — DataSF. (n.d.-b). <https://data.sfgov.org/Energy-and-Environment/Elevation-Contours/rnbg-2qwx>
8. FastStats. (n.d.). Body Measurements. <https://www.cdc.gov/nchs/fastats/body-measurements.htm>
9. Google Maps. (n.d.). Google Maps. <https://www.google.com/maps>
10. Hara, T. (1965). Method of measuring the aerodynamic drag of trains. *Bulletin of JSME*, 8(31), 390–396. <https://doi.org/10.1299/jsme1958.8.390>
11. Indot. (2022, September 12). Railroad crossing facts. INDOT. <https://www.in.gov/indot/safety/traffic-safety/railroad-crossing-facts/>
12. Khodaparastan, M. (2020). Recuperation of Regenerative Braking Energy in Electric Rail Transit Systems. Dissertations: City College of New York. https://academicworks.cuny.edu/cgi/viewcontent.cgi?params=/context/cc_etds_theses/article/1935/&path_info=Disertation_Regenerative_Braking_Energy_in_Electric_Rail_Transit_System_AM.pdf
13. Maanshan Kingrail Technology Co.,Ltd [Tyler]. (2024, June 1). Rolling Resistance of Railway Wheels. Kingrail. Retrieved August 3, 2025, from <https://www.kingrailparts.com/Blog/217.html>
14. Metropolitan Transportation Authority. (2003b). REGULAR BOARD MEETING DECEMBER 4, 2003. https://libraryarchives.metro.net/board/Items/2003/12_December/20031126Item16.pdf
15. Panchenko, S., Gerlici, J., Lovska, A., & Ravlyuk, V. (2025). Assessment of uneven wear of freight wagon brake pads. *Applied Sciences*, 15(12), 6860. <https://doi.org/10.3390/app15126860>
16. Point-to-Point versus Hub-and-Spoke Networks — The Geography of Transport Systems. (2023, June 12). The Geography of Transport Systems — the Spatial Organization of Transportation and Mobility. <https://transportgeography.org/contents/chapter2/geography-of-transportation-networks/point-to-point-versus-hub-and-spoke-network/>
17. Powerful route planner that prefers greenery and can generate round trip routes of a specified distance — Trail Router. (n.d.-b). Trail Router. <https://trailrouter.com/>
18. SFMTA’s Transit Economic Benefits Study. (2015). In www.sfmta.com. San Francisco Munipal Transportation Agency. Retrieved August 8, 2025, from <https://www.sfmta.com/sites/default/files/agendaitems/2015/9-3-15>
19. Siemens Mobility, Inc. (2020). Muni LRV4 High-Floor Light Rail Vehicle [Report]. <https://assets.new.siemens.com/siemens/assets/api/uuid:02801dad-5a01-478c-beb0-8ef209289802/SIE-RS-SanFran-S200-Sheet-10-9-19-v2.pdf>
20. Udriste, C., Tevy, I., & Antonescu, P. (2023). Optimal control problem for minimization of net energy consumption at Metro. *Mathematics*, 11(4), 1035. <https://doi.org/10.3390/math11041035>
21. Vision Zero Benchmarking: Commute Methods — SF.gov. (n.d.). <https://www.sf.gov/data--vision-zero-benchmarking-commute-methods>
22. Wikipedia contributors. (2025a, March 13). Drag equation. Wikipedia. https://en.wikipedia.org/wiki/Drag_equation
23. Wikipedia contributors. (2025b, July 30). Muni metro. Wikipedia. https://en.wikipedia.org/wiki/Muni_Metro

¹ChatGPT was used to refactor my code and clean up comments. All code is my own authorship.

24. Wikipedia contributors. (2025c, August 2). J Church. Wikipedia. https://en.wikipedia.org/wiki/J_Church

6. APPENDIX

6.1. Calculation of Brake Wear Cost. The first step in the calculation is computing the braking duration. Using Indiana's government website, we use their values that say it takes a train a train going 55 miles per hour 1.5 miles to stop [11]. We convert these values to m s^{-1} and m (24.59 m s^{-1} and 2414 m). Assuming that braking is constant, we plug into the equation for constant acceleration to get the braking duration,

$$a = \frac{v^2}{2d} = \frac{24.59^2}{2 * 2414.01} = 196.36 \text{ s}.$$

From another source [15], we get the average brake force to be $35,000 \text{ N}$, and that 35% of brake wear corresponds to $74,400 \text{ km}$ of stopping. The lifetime of a brake is then $\frac{74400}{0.35} = 212571.43 \text{ km}$. We know the stop spacing is 2.4 km , so we divide the value to get $88,571.43$ stops for the lifetime of a brake. Now we can compute the total impulse over the lifetime of a brake:

$$88,571.43 * 35,000 * 197 = 6.11 \times 10^{11} \text{ N} \cdot \text{s}$$

We divide this by the cost of a brake [14] to get dollars per N.

$$\frac{2,464.79}{6.11 \times 10^{11}} = 4.04 \times 10^{-9} \frac{\$}{\text{N} \cdot \text{s}}$$

6.2. Table of Arrival Times.

Smoothing Parameter Table

λ_{smooth}	Time (min)	Total Cost
0.0001	1.05	10947.75
0.0010	1.67	30129.01
0.0100	2.71	54937.50
0.1000	4.34	91400.38

6.3. Passenger Fraction Function.

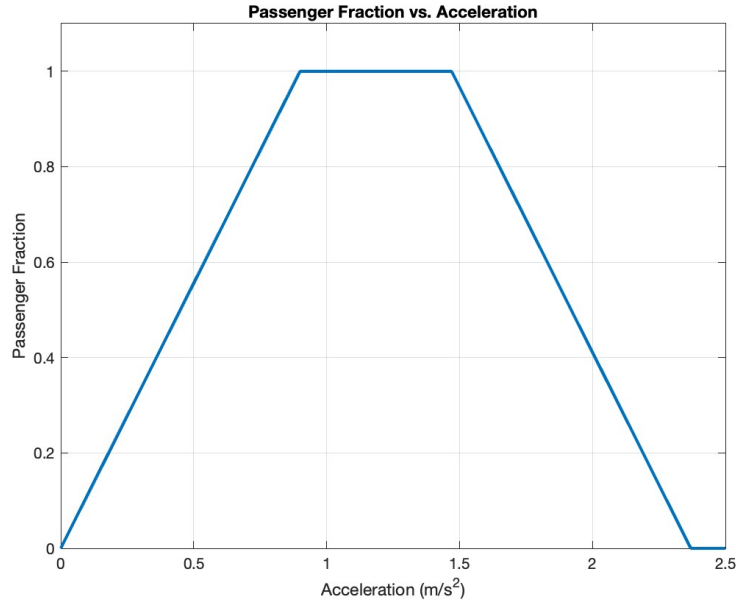


FIGURE 7. Optimal Trajectories for Different Hills